

ARYAN MAHESHWARI

aryan.maheshwariTX@gmail.com | (281) 763-9846 | Houston, Texas
aryanmaheshwaritx-ux.github.io/internships

EXPERIENCE

Founder & Lead Engineer | May 2025 - Present

Aero Robotics (Student Venture)

- Filed a USPTO patent application (Sept. 2025) for a wind turbine blade maintenance payload; designed and built a system for surface preparation, composite-safe cleaning, adhesive application, and precision sensor placement within payload weight limits.
- Built a drone-deployed robotic system for hazardous industrial environments, with wind turbine blade inspection as the first target application; development is ongoing.
- Prototyped a modular quick-swap drone attachment platform in Fusion 360; iterated 3D-printed designs for vibration isolation, structural stability, and compatibility across drone platforms.
- Piloted 20 field deployments at 125 m for a commercial wind energy client in Minnesota; pursuing Part 107 certification.

Commercial Operations Intern | Jun 2024 - Aug 2024

Mars Materials (ClimateTech, backed by Breakthrough Energy)

- Prepared investment decks, reviewed investor theses, and drafted investor outreach directly with the CEO to support fundraising.
- Turned post-run technical debriefs from the pilot system at Shell Technology Center into reusable operational workflows.
- Contributed to grant applications and designed elements of an internal employee recognition program.

Product Design Intern | May 2023 - Aug 2023

Predyct (ClimateTech) | predyct.io

- Informed the core sensor specification through research on wind turbine blade failure statistics and low-power sensing requirements.
- Identified market gaps through competitive analysis of blade monitoring solutions, helping shape product direction.

EDUCATION

University of Illinois Urbana-Champaign

B.S. Aerospace Engineering | Minor in Business | Expected May 2030

Lawrence E. Elkins High School, Engineering Academy

Honors Diploma, 2026 | Engineering & Computer Science

TECHNICAL SKILLS

Flight Systems: ArduPilot, Pixhawk-class flight controllers, GNSS/RTK integration, ESC and radio systems, battery and payload budgeting

Mechanical Design: Siemens NX, Autodesk Fusion 360 (Certified), 3D prototyping, payload structural design, mounting systems

Electronics: Soldering, circuit integration, sensor systems, connector and wiring selection

Programming: Python, Java, HTML/CSS | **Other:** AI/ML fundamentals, investment deck preparation

ENGINEERING CLUBS & SELECTED ACTIVITIES

- **Illini Electric Motorsports** | Member
- **Liquid Rocketry** | Member
- NASA High School Aerospace Scholars (HAS) Program | Oct 2024 - Jul 2025
- TSA Drone Challenge | Pilot, State Finalist (2026)
- TEX-E ClimateTech Entrepreneurship Bootcamp | Apr 2025
- VEX Robotics Competition | District Award (2023)
- Chinmaya Mission Houston | Student Teacher & Cultural Volunteer (2018 - Present)